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Acronics Solutions

NETWORK SECURITY EQUIPMENT
High speed Anti-DDoS function

OUTSTANDING FEATURES

Easy system integration

Easily deployed in Server and Data Center systems (between Router and Firewall).

Comprehensive solution with high performance

Provides a complete solution to detect and mitigate Multi-layer DDoS attacks at Layer 2/3 including bandwidth attacks, protocol attacks and Layer 6/7 application layer attacks.

Using FPGA acceleration hardware helps DDoS Defender handle high-intensity attacks at high speed, provide early warnings and effectively prevent attacks.

Optimal management and monitoring system

The tracking system provides complete information to help administrators easily manage and handle attacks.

Detect and warn administrators of attacks via email.

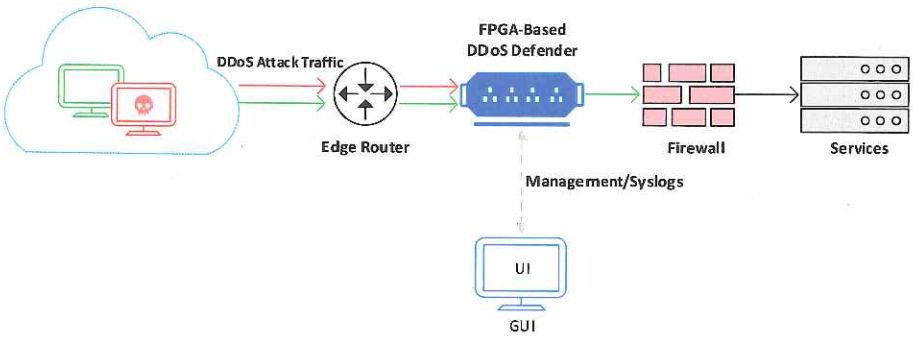
NECESSITY OF DDOS DEFENDER SOLUTION

Nowadays, DDoS attacks are increasingly large in scale and diverse in attack forms. The development of IoT and DDoS attack tools is a powerful resource for attackers to create DDoS attacks with intensity up to hundreds of Gbps. Attacks are increasingly complex with the creation of new attack vectors and combining vectors together (multi-vector), making detection and prevention of DDoS increasingly difficult.

This leads to an urgent need for a product that meets the increasing variability and intensity of DDoS attacks.

DDOS DEFENDER SOLUTION

FPGA/ASIC-Based DDoS Defender, using FPGA/ASIC technology with parallel hardware processing capabilities, is an effective solution in detecting and minimizing DDoS attacks at the network edge with the ability to detect attacks. attack early as well as respond to large bandwidth attacks.



Main features:

Detect DDoS attacks:

- Attack detection is based on thresholds and a set of packet attributes for each IP to detect unusual signs and thereby identify attacks.
- Types of detected attacks: Bandwidth attacks: UDP/ICMP Flood,...; Protocol attack: SYN Flood,..; Application layer attacks: HTTP Flood, HTTPS Flood, Slowloris,..
- Attack detection time is less than 1 second.

Prevent DDoS attacks:

- Implement Challenge-Response mechanism to block attack packets and ensure service to valid clients; Bandwidth limit.
- Build a White list/Black list to drop packets from suspected attack sources.

Tracking DDoS attacks:

- Track ongoing attacks: Attack details (IP information, bandwidth information, attack type)

Appliance

Device	
FPGA	Xilinx / Altera FPGA
Processor	ARMs / Intel Processors
Inspection port	
10/100/1000 Copper Ethernet	2; 4; 6; 8 Ports
10G Fibe-roptic SFP/SFP+	2; 4 Ports
Management port	
10/100/1000 Ethernet or Fiber Optic or UART	1

Protection Features

Performance	
Max Mitigation Capacity Throughput	8; 10; 12 Gbps
Max Legit Concurrent Sessions	3,000,000 – 6,000,000
Max Attack Concurrent Sessions	3,000,000 - – 6,000,000
Max DDoS Flood Attack Prevention Rate	10,000,000 pps
Latency	< 60 microseconds
SSL/TLS Connections per Second	20KCPS (RSA 2K)
Real-Time Signatures	Detect attacks and protect in less than 1 second
Operation mode	
Network Operation	Transparent L2 Forwarding/IP Forwarding
Deployment Modes	Inline active; inline inactive (reporting, no blocking); SPAN port monitor
IPv6	Yes
Jumbo Frame	Supported
DDoS Protection	Volumetric/Protocol/Application attack protection. TCP/UDP/ICMP/HTTP/HTTPS flood attacks, TCP/UDP Fragmentation attacks, Spoof detection, Amplification attack, Synonymous IP attack, Slow-rate HTTP attack.
Block Actions	Block attacks in real time with: Dynamic Filters, Challenge-Response for TCP, HTTP and DNS suspicious traffic, verify HTTPS Access, Managing SSL/TLS Certificates, HTTP Authentication, Proxy-based SYN Cookies, TCP Connection Resetting, Add to white/black list, Connection Blocking, Dynamic Source IP Blocking, Connection Rate Limit.
Management	
Management	SNMP v1; Web UI
Reporting	Real-time and historical traffic reporting, drill-down by attack types, blocked sources, top sources by IP location
Supported Browsers	Internet Explorer, Firefox, Chrome
Notifications	SNMP trap, Log file, Syslog